

Compensation for Psychological and Physical Occupational Disamenities: A Multiple Linear
Regression Analysis of U.S. Workers, 2006–2024

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Abstract:

This paper examines whether psychological disamenities command compensating wage differentials in the U.S. labour market and how these premiums vary across demographic and institutional contexts. Using O*NET 30.0 measures of occupational disamenities merged with CPS MORG microdata from 2006 to 2024 ($N = 58,070$), we estimate the relationship between stress tolerance and hourly wages. We find a U-shaped association: wages decline at moderate levels, but rise at extreme levels, suggesting that workers are only compensated for severe burdens. We further find that institutional and social bargaining power affect compensation. Institutionally, unionization is associated with approximately 14% higher wages, but does not change the marginal compensation for either psychological or physical disamenities. Socially, women and visible minority workers receive systematically lower premiums, while Black workers experience slightly higher marginal returns. Finally, we consider several limitations to our data that restrict any causal interpretation of discrimination. Overall, we find that observed wage premiums for occupational disamenities differ significantly across workers' visible identities and institutional contexts.

Introduction & Literature:

The basic compensating wage differentials model states that wages reflect not only productivity, but also compensation for non-wage amenities/disamenities, such as pay bonuses, risks to life, or the general unpleasantness of specific work (Rosen, 1986). For example, to accept a job that poses a risk to their health, a worker must receive higher wages or benefits; otherwise, they would opt for a safer job with the same pay to maximize their utility. Substantial literature supports the existence of wage premiums for physical disamenities. However, as advanced economies deindustrialize, psychological risk may replace physical risk as the dominant form of workplace disamenity, for which much less is known.

Theoretically, physical and psychological disamenities should be equally compensated through wages, yet empirical research has primarily focused on physical compensating wage differentials (Thaler & Rosen, 1975; Smith, 1974). Evidence for psychological disamenities remains ambiguous. Some studies identify significant premiums for stressful jobs where workers are willing to forgo almost 10% of their wage to avoid frequent tight deadlines (French & Dunlap, 1997; Nagler et al., 2023). Others find small to negligible premiums in frictional labour markets, such as government or youth-concentrated occupations, where prohibitively high search costs or informational asymmetries cause compensating differentials to break down (Bonhomme & Jolivet, 2009; Nagler et al., 2023). Psychological compensating wage differentials may also vary systematically by gender and bargaining power. Women often report higher levels of work-related stress and burnout than men (Padkapayeva et. al, 2018; Artz B et. al, 2022), and experimental research shows that women are willing to accept lower wages for job flexibility and security, while men prioritize wage growth and prestige (Wiswall & Zafar, 2018). Studies measuring physical risks in union vs non-union industrial sectors find that, given greater worker bargaining power, base wages will converge to the true cost of occupational disamenity (Fairris, 1992). The lack of consensus on this topic suggests that the existing literature underestimates the scale and variability of psychological disamenities.

The Context and Data:

Our analysis uses pooled cross-sectional data of 58,070 individual workers constructed by merging occupational data from O*NET 30.0 with NBER extractions of the CPS Annual Earnings File (MORG) from 2006-2024. The merge is performed at the SOC-occupational level

using standardized O*NET-SOC 2019 crosswalks to ensure that, as the occupational classification system changes across time, the underlying occupations remain constant. Our dependent variable is real hourly wages, and our unit of observation is a worker's household. CPS survey data are independently and identically drawn from the U.S. population, and nominal income observations are truncated at \$99.9/hour to prevent outliers. We further use the NBER's included analytical earnings weights (earnwt) to produce nationally representative estimates of the U.S. labour force. O*NET provides measures of stress tolerance, physical work context, and work values, and CPS survey data provides weighted household microdata for demographics, labour market outcomes, and hourly real wages (2023 base-year).

Stress is a heterogeneous, multidimensional measure, and self-reported psychological stress suffers from significant "measurement bias due to differences in respondent interpretations and perceptions" (French & Dunlap, 1997). To reduce this bias, we use standardized expert-assessed O*NET measures instead of subjective worker reports. Stress tolerance is a continuous measure of occupational-specific psychological pressure that is assessed by O*NET experts on a 1 ("least important") to 5 ("most important") scale and ranges from 2.77 to 4.95, with a mean of 4.15, a median of 4.23, and an IQR of 0.63. Work context aggregates exposure to twelve physical disamenities (e.g., noise, hazards, heights), each rated from 1 ("never") to 5 ("every day") and ranges from 1.05 to 4.03, with a mean of 1.98, a median of 1.82, and an IQR of 0.88. Work values are included to control for intrinsic ("Achievement") and extrinsic ("Recognition") rewards (1-7 extent scale), which are forms of non-wage compensation that introduce endogeneity bias and confound the relationship between stress and wages.

Table 1 reports descriptive statistics. A representative worker in the data is approximately a 40.8-year-old, non-unionized male or female with some college education, working a moderately stressful white collar job, earning \$22.64/hour. Approximately 11% of workers are unionized or covered by a union contract, and the wage distribution is right-skewed.

Variable	Mean	Median	Std. Dev.	Min	Max	P25	P75	IQR
wage_2023	22.64	19.28	12.46	0.11	130.76	14.62	26.98	12.36
stress_tolerance	4.15	4.23	0.42	2.77	4.95	3.85	4.48	0.63
work_context	1.98	1.82	0.62	1.05	4.03	1.49	2.37	0.88
work_values	4.16	4.26	1.19	1.33	6.67	3.34	5.17	1.83
age	40.8	40	15.09	16	85	28	53	25
grade92	40.07	40	2.35	31	46	39	42	3
union_any	0.11	0	0.31	0	1			

(Table 1)

Regression Specification:

To evaluate compensating wage differentials, we estimate the following model:

$$\begin{aligned} \ln(wage_i) = & \beta_0 + \beta_1 work_context_i + \beta_2 work_context_i^2 + \beta_3 work_values_i + \beta_4 stress_tolerance_i + \beta_5 stress_tolerance_i^2 \\ & + \beta_6 female_i + \sum_r \beta_{7r} race_{ir} + \beta_8 age_i + \beta_9 age_i^2 + \sum_g \beta_{10g} grade_{ig} + \sum_c \beta_{11c} class_worker_{ic} + \beta_{12} union_i \\ & + \beta_{13}(stress_tolerance_i \times union_i) + \beta_{14}(work_context_i \times union_i) + \beta_{15}(stress_tolerance_i \times female_i) \\ & + \sum_r \beta_{16r}(stress_tolerance_i \times race_{ir}) + \sum_c \beta_{17c}(stress_tolerance_i \times class_{ic}) \\ & + \sum_t \delta_t year_{it} + \sum_j \theta_j industry_{ij} + \sum_s \phi_s state_{is} + \varepsilon_i \end{aligned}$$

We transform the dependent variable (hourly wage) by the natural logarithm and use robust standard errors to correct for right-skewed earnings data and heteroskedasticity. Quadratic terms account for the nonlinear association between occupational disamenities and wages. To examine heterogeneity compensation by demographic and institutional contexts, we further control for sex, race, education, employer organization type (e.g., private vs. public), and unionization, as well as interactions between stress tolerance and these characteristics. The included fixed effects are jointly statistically significant and control for unobserved wage variation to allow our model to explain within-year, within-state, and within-industry occupational variation rather than between-year, between-state, and between-industry occupational variation. In other words, the model compares workers in similar occupations across industries and locations at the same time, rather than making an unfair comparison between an accountant on Wall Street in 2024 and an accountant in the retail industry in 2010, living in Idaho.

Regression Analysis:

Table 2 presents six specifications of the estimated relationship between stress and wages across 58,070 pooled cross-sectional worker-households from 2006 to 2024. Specification (1) includes only stress tolerance and produces a precise but economically modest negative estimate, indicating that stress tolerance alone has little meaningful effect on wages. Moving from the 25th to the 75th percentile of stress is associated with an approximate 7% decrease in wages. Specifications (2)-(6) sequentially introduce nonlinear terms as well as demographic, institutional, and fixed effect controls to better explain how occupational stress affects compensation across workers.

In specification (3), both stress tolerance and exposure to physical disamenities have negative linear coefficients and positive quadratic coefficients, implying a U-shaped relationship between disamenities and wages. This U-shape is the result of high-wage jobs clustering at opposite ends of the disamenity range: low-psychological but high-physical disamenity jobs (e.g., trades) and high-psychological but low-physical disamenity jobs (e.g., executives or managers). As disamenities increase, wages initially fall, and then begin to increase only after a high threshold, at which point the market begins compensating workers for the disutility of their occupational disamenities.

The turning point for physical disamenities is approximately 2.4, and for psychological disamenities it is approximately 4, both on a 1-5 scale. The stress turning point (≈ 4) lies near the 70th percentile, meaning that only workers in the upper third of the stress distribution are compensated for more stressful work. The hazard turning point (≈ 2.4) lies near the 75th percentile of the physical exposure distribution, so only workers in the upper quartile are compensated for more dangerous work. These results suggest that (1) compensating wage differentials operate efficiently only at extreme levels of disamenity, and (2) the negative returns imply that workers are willing to accept lower wages to avoid stressful or hazardous occupations, a finding consistent with the literature that many workers forgo earnings in exchange for more comfortable work environments.

Specification (4) improves on (3) because it includes worker identities and finds that disamenity premiums vary significantly across demographic groups. This inclusion greatly increases within R-squared (i.e., remaining variation explained after fixed effects) from 0.021 to 0.167, indicating that demographic characteristics explain a large share of the previously unaccounted-for variation in wages across workers. Figure 1 plots the findings from specification (4) and shows that men receive higher marginal returns to stress across the entire domain of occupational disamenities. The female indicator coefficient is positive (+0.139, $p < 0.01$), while the interaction term Female $\times$ Stress Tolerance is negative (-0.057, $p < 0.01$). The positive female indicator coefficient should not be interpreted as women earning more than men in low-stress occupations, since the observed domain of stress does not include those extremely low values. Compared to men, women's stress tolerance must be 0.4 points (8.8%) higher for the market to begin compensating stress equivalently, and even after this delayed turning point, their marginal return remains significantly lower.

Specifications (5) and (6) improve on (4) by including institutional contexts and interactions between race and stress compensation. Figure 2 shows that, consistent with classical findings, unionization is associated with a substantial 15.7% or approximately \$4.72 per hour wage premium. This premium confirms what economic theory predicts, that greater collective bargaining power reduces employer monopsony power and drives wages up. However, unionized workers do not earn more as stress increases ($\text{Union} \times \text{Stress Tolerance} = -0.001, p > 0.10$) or as hazard exposure increases ($\text{Union} \times \text{Exposure} = 0.008, p > 0.10$), suggesting that unions do not materially affect the marginal compensation for occupational disamenities. This finding does not contradict our hypothesis that compensation for occupational disamenities depends on bargaining power. As Fairris (1992) notes, hazardous or stressful occupations often unionize because workers seek protection against these very disamenities (p. 215). Consequently, the positive association between unions and disamenities may reflect self-selection rather than a causal effect of unions on compensation. The precise null effects we estimate for marginal disamenity compensation may reflect the difficulty unions face in drafting and enforcing collective bargaining agreements for specific occupational disamenities. Instead, unions may negotiate compensation for the total disamenity burden and include these costs as higher average wages rather than marginal returns. We also find that working for the federal government is associated with approximately 10% higher wages relative to all other employer types, but (similar to unionization) does not affect the marginal compensation for occupational disamenities.

We test whether all race-stress interaction terms are jointly zero, which produces a large F-test ($F = 2.015, p = .0016$). We reject the null and infer the alternative hypothesis that there is heterogeneous marginal compensation by race. Figure 3 further shows significant racial variation in both base wages and marginal returns to stress. Asian workers earn the highest wages at low stress levels, but experience negative marginal compensation for stress ($\text{Asian} \times \text{Stress Tolerance} = -0.061, p < 0.01$). White workers (omitted category to avoid the dummy variable trap) earn the highest wages at high stress levels. Black workers start at a lower base wage ($\text{Black} = -0.173, p < 0.01$) but exhibit a modest positive marginal compensation for stress ($\text{Black} \times \text{Stress Tolerance} = +0.022, p \approx 0.09$). Conditional on working a stressful job, stress-compensating wage differentials appear efficient and beneficial for Black workers, but inefficient or even punitive for Asian workers.

Figure 4 shows that across all racial groups, gender-based differences dominate race-based differences and systematically lower the marginal return to stress. The cumulative disadvantage is strongest for Asian women who exhibit the flattest and lowest psychological compensation curves at high stress levels.

Limitations to research:

Our results are subject to three primary limitations: (1) the limitations of cross-sectional data for identifying discrimination, (2) external invalidity in sufficiently different labour markets, and (3) omitted variable bias from unobserved worker characteristics.

First, because CPS data are cross-sectional, wage differences and differences in marginal stress compensation across demographic groups cannot be interpreted as evidence of discrimination. Because stress tolerance does not vary within occupations, including occupation fixed effects would induce perfect collinearity and remove all useful identifying variation, a violation of LSA.4 ($\text{Var}[X] \neq 0$). Although we control for observable characteristics, we cannot separate discrimination from self-selection without a causal identification strategy (e.g., IV, pseudo-experimental). As a result, our findings remain descriptive.

Second, external validity is limited to labour markets similar to the United States. While it may be reasonable to extend these findings to Canada or other OECD economies due to similarities in regulatory environments, industry composition, and levels of human capital, they should not be extended to labour markets in markedly different contexts, as the results cannot be assumed to operate similarly.

Third, CPS omits many determinants of wages and job sorting, which are threats to internal validity. This limitation reflects a violation of LSA.1 ($E[u|X] = 0$), because unobserved traits, such as cognitive ability or risk preferences, are correlated with both choice of occupation and, consequently, wages. We improve this by controlling for intrinsic and extrinsic non-wage compensation to proxy these unobserved traits, but direct measures remain omitted. Finally, these CPS data exclude individuals who are out of the labour force, including those who may have exited because compensation for stressful work is below their reservation price. These workers (e.g., retirees or part-time workers) might have otherwise entered the labour force or worked more if stress were better compensated.

Conclusion:

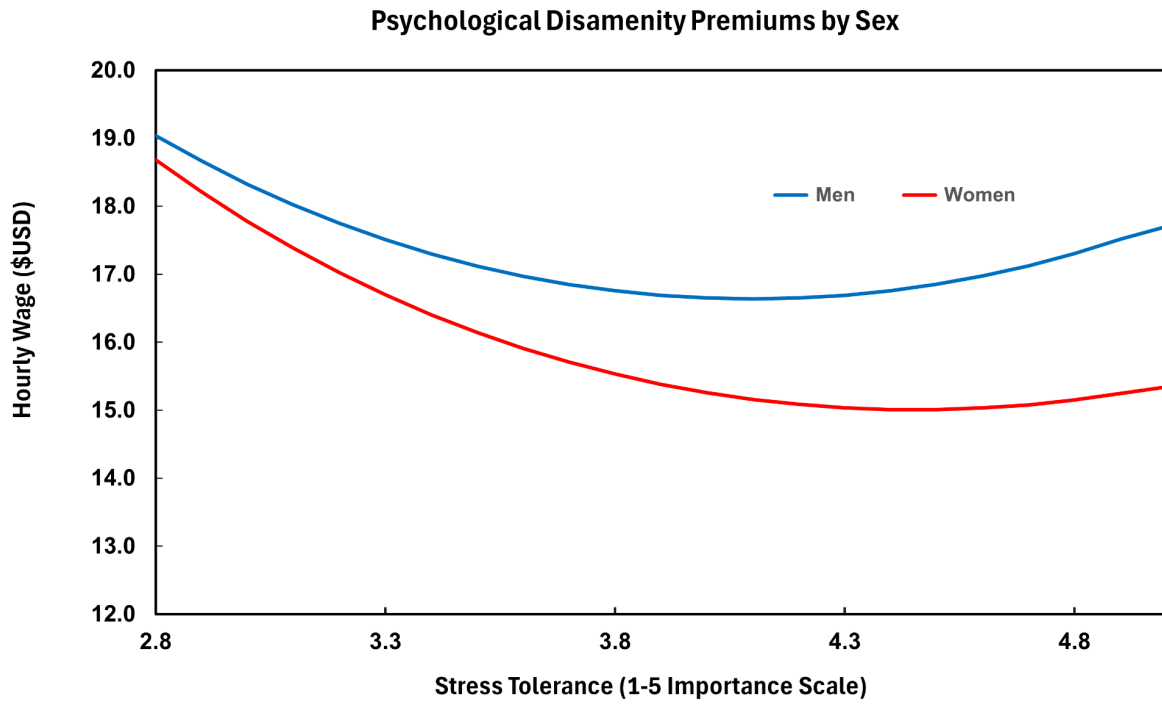
In this paper, we found that compensating wage differentials for psychological occupational disamenities exist, but they are socially stratified and unevenly distributed. Psychological compensating differentials appear efficient only at extreme levels and primarily for groups with greater social or institutional bargaining power. Although unions are associated with higher average wages, consistent with their improved bargaining position, unions are not associated with differences in marginal returns to either the psychological or physical costs of work. Similarly, employer organization type (public/private) shows no significant association with compensating wage differentials. The psychological costs of work are therefore unequally priced across demographic and institutional contexts. Further research should use causal methods to determine whether sorting or discrimination causes our observed differences in compensation for psychological disamenities.

Compensation for Psychological and Physical Occupational Disamenities: A Multiple Linear Regression Analysis of U.S. Workers, 2006–2024

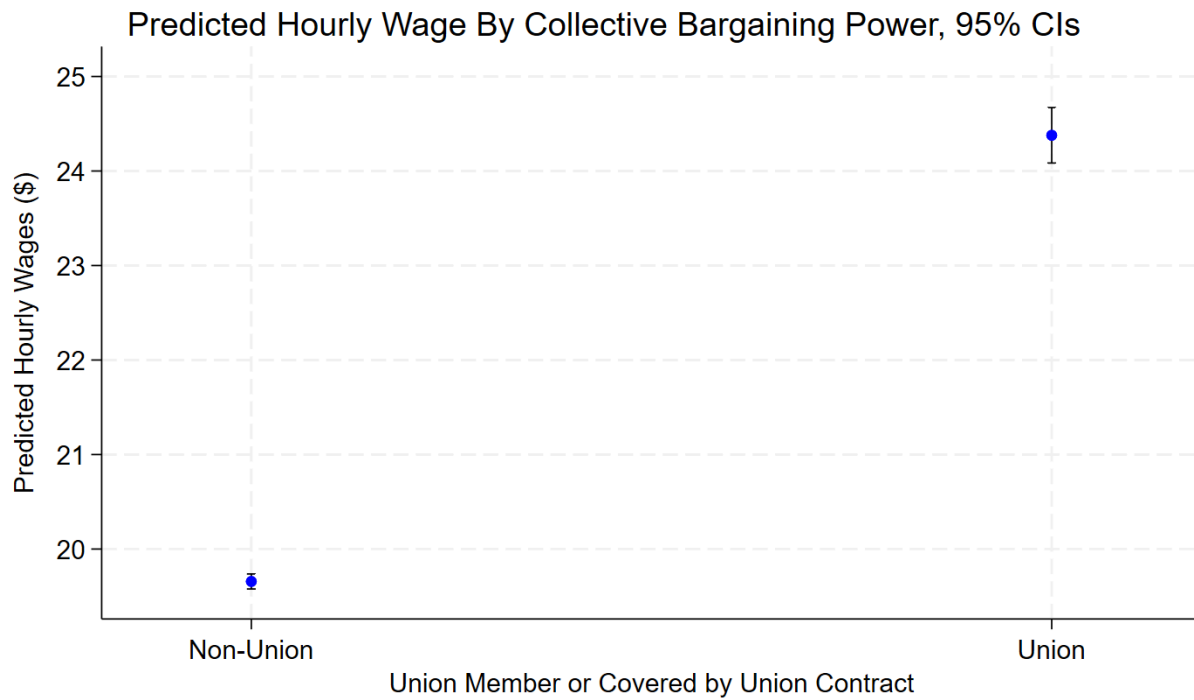
Variables	(1) ln (Wage)	(2) ln (Wage)	(3) ln (Wage)	(4) ln (Wage)	(5) ln (Wage)	(6) ln (Wage)
Stress Tolerance	-0.111*** (0.005)	-0.123*** (0.006)	-1.033*** (0.086)	-1.050*** (0.084)	-0.686*** (0.078)	-0.651*** (0.079)
Exposure to Physical Disamenities		-0.130*** (0.004)	-0.713*** (0.031)	-0.660*** (0.031)	-0.481*** (0.028)	-0.481*** (0.028)
Recognition and Achievement Work Values		-0.034*** (0.002)	0.018*** (0.002)	0.021*** (0.002)	0.013*** (0.002)	0.012*** (0.002)
Stress Tolerance Squared			0.122*** (0.011)	0.125*** (0.010)	0.081*** (0.010)	0.080*** (0.010)
Exposure to Physical Disamenities Squared			0.149*** (0.007)	0.136*** (0.007)	0.099*** (0.006)	0.099*** (0.006)
Female				-0.092*** (0.005)	-0.097*** (0.004)	0.139*** (0.039)
Black				-0.083*** (0.006)	-0.084*** (0.006)	-0.173*** (0.053)
Asian				0.038*** (0.011)	-0.018* (0.010)	0.231*** (0.089)
Age					0.028*** (0.001)	0.028*** (0.001)
High School (GED)					0.105*** (0.036)	0.107*** (0.036)
Bachelor's Degree					0.296*** (0.037)	0.297*** (0.037)
Professional Degree (JD, MD)					0.507*** (0.050)	0.509*** (0.050)
Working for State Government					-0.110*** (0.018)	-0.110*** (0.018)
Working for Local Government					-0.105*** (0.016)	-0.103*** (0.016)
Working for Private, For Profit					-0.101*** (0.016)	-0.101*** (0.016)
Working for Private, Non-Profit					-0.086*** (0.018)	-0.087*** (0.018)
Union Member or Covered by Union Contract					0.147*** (0.006)	0.136*** (0.064)
Union × Stress Tolerance						-0.001 (0.014)
Union × Exposure to Physical Disamenities						0.008 (0.009)
Female × Stress Tolerance						-0.057*** (0.009)
Black × Stress Tolerance						0.022* (0.013)
Asian × Stress Tolerance						-0.061*** (0.021)
Constant	3.448*** (0.021)	3.895*** (0.025)	5.837*** (0.181)	5.864*** (0.178)	4.250*** (0.170)	4.136*** (0.173)
Industry, Year, State Fixed Effects	NO	NO	YES	YES	YES	YES
Observations	58,070	57,917	57,917	57,917	57,917	57,917
R-squared	0.009	0.033	0.286	0.392	0.393	0.393
Adjusted R-squared	0.009	0.033	0.284	0.390	0.391	0.391
Within R-squared	0.009	0.033	0.021	0.167	0.168	0.168

Notes: Robust standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01

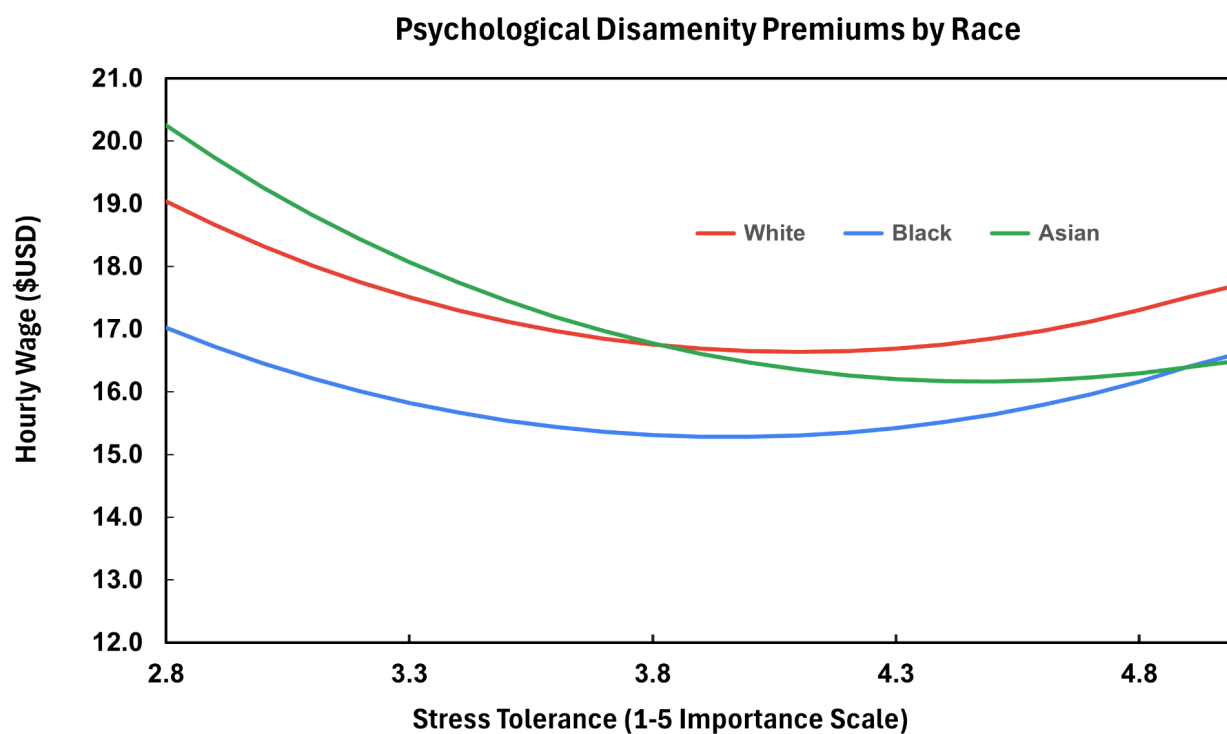
(Table 2)



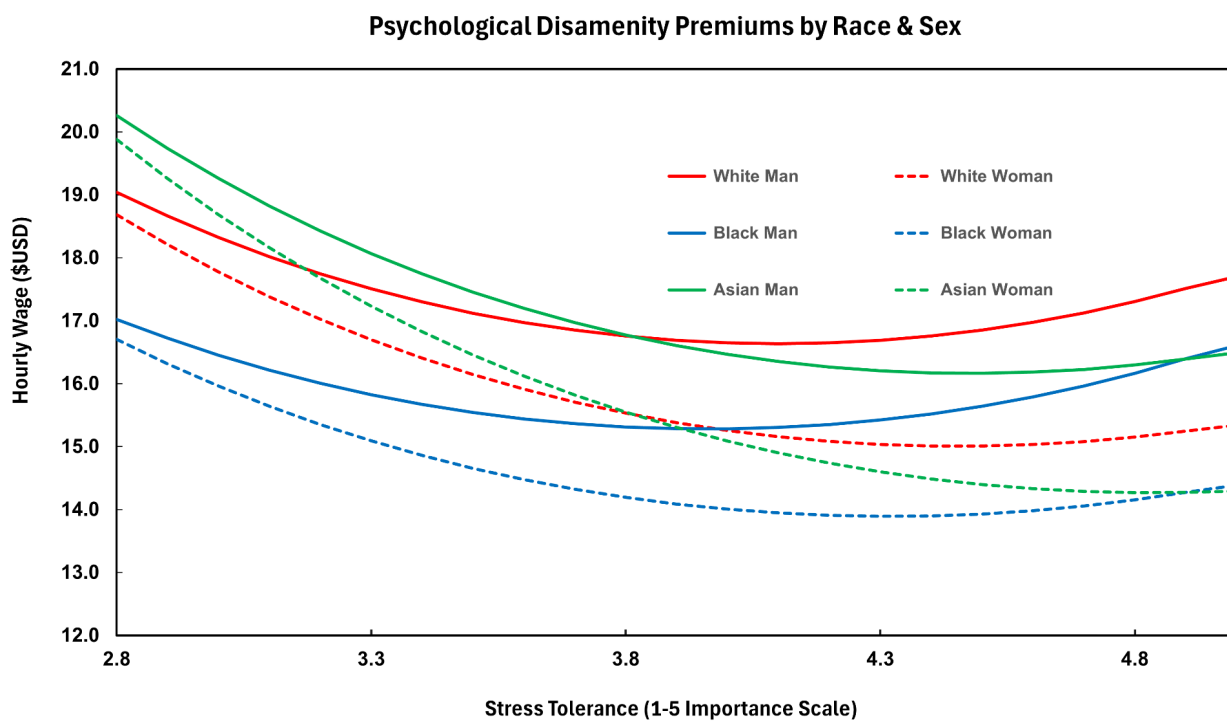
(Figure 1)



(Figure 2)



(Figure 3)



(Figure 4)

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